Rates of Change and Rectilinear Motion

ANSWER KEY

Section 1: Position, Velocity & Speed

1 A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. At what times is the particle at rest?

a. $t = 1$ **and** $t = 3$ ✓

b. $t = 0$ and $t = 3$

c. $t = 2$

d. $t = 3$ only

At rest means velocity $v(t) = x'(t) = 0$: $3t^2 - 12t + 9 = 0 \Rightarrow 3(t-1)(t-3) = 0 \Rightarrow t = 1, 3$.

2 What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

a. 8 ✓

b. -8

c. 5

d. $\sqrt{73}$

Speed is defined as the absolute value of velocity: $|v(t)| = |-8| = 8$.

3 When is a particle's speed increasing?

a. **When velocity and acceleration have the same sign** ✓

b. When velocity is positive

c. When acceleration is positive

d. When acceleration is zero

Speed $|v(t)|$ increases when velocity and acceleration work in the same direction, meaning $v(t)$ and $a(t)$ have the same sign.

Section 2: Total Distance vs. Displacement

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4 If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

a. -3 ✓

b. 5

c. -4

d. 3

Displacement $= s(3) - s(0) = (9 - 12) - 0 = -3$.

5 For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

a. 5 ✓

b. 3

c. 4

d. 7

$v(t) = 2t - 4 = 0 \Rightarrow t = 2$. From $t = 0$ to 2: $|s(2) - s(0)| = |-4 - 0| = 4$. From $t = 2$ to 3: $|s(3) - s(2)| = |-3 - (-4)| = 1$. Total distance $= 4 + 1 = 5$.

6 Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

a. $\int_a^b |v(t)|dt$ ✓

b. $\int_a^b v(t)dt$

c. $|\int_a^b v(t)dt|$

d. $\int_a^b a(t)dt$

Total distance accumulates the speed over time: $\int_a^b |v(t)|dt$.

Section 3: Related Rates: Geometric Models

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7 The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π **cm^2/s ✓**

b. 30π cm^2/s

c. 100π cm^2/s

d. 20π cm^2/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm^2/s.

8 A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?

a. $-\frac{5}{6}$ **ft/s ✓**

b. $-\frac{12}{5}$ ft/s

c. -2 ft/s

d. $-\frac{5}{12}$ ft/s

$x^2 + y^2 = 13^2 = 169$. When $x = 5$, $y = 12$. Differentiating:
$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow 5(2) + 12\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{10}{12} = -\frac{5}{6}$ ft/s.

9 Air is pumped into a spherical balloon at a rate of 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ **cm/s ✓**

b. $\frac{1}{4\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 25π cm/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt} \Rightarrow 100 = 4\pi(25)\frac{dr}{dt} \Rightarrow 100 = 100\pi\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi}$ cm/s.

Section 4: Related Rates: Angles & Proportions

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10 A conical tank with vertex down has height 10 m and base radius 4 m. Water flows in at 2 m^3/min. What is the relation between r and h?

a. $r = \frac{2}{5}h$ ✓

b. $r = \frac{5}{2}h$

c. $r = \frac{1}{2}h$

d. $r = 4h$

By similar triangles in the cross-section: $\frac{r}{h} = \frac{4}{10} = \frac{2}{5} \Rightarrow r = \frac{2}{5}h$.

11 Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m^3/min?

a. $\frac{1}{2\pi}$ **m/min** ✓

b. $\frac{2}{\pi}$ m/min

c. $\frac{1}{4\pi}$ m/min

d. $\frac{4}{25\pi}$ m/min

$\frac{dV}{dt} = \frac{4}{25}\pi h^2 \frac{dh}{dt} \Rightarrow 2 = \frac{4}{25}\pi(25)\frac{dh}{dt} = 4\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{4\pi} = \frac{1}{2\pi}$ m/min.

12 A street light is mounted at the top of a 15-foot pole. A 6-foot man walks away from the pole at 5 ft/s. How fast is the tip of his shadow moving?

a. $\frac{25}{3}$ **ft/s** ✓

b. 5 ft/s

c. $\frac{10}{3}$ ft/s

d. 10 ft/s

Let s be shadow tip distance: $\frac{s}{15} = \frac{s-x}{6} \Rightarrow 6s = 15s - 15x \Rightarrow 9s = 15x \Rightarrow s = \frac{5}{3}x$. Then $\frac{ds}{dt} = \frac{5}{3}\frac{dx}{dt} = \frac{5}{3}(5) = \frac{25}{3}$ ft/s.

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